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Structure and Biomedical Properties of Lignins (A Review)

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Abstract—This literature review is in line with current trends in the development of nature-like technologies and is devoted to the analysis of the chemical structure and to the search for new ways to use lignins for biomedical applications. Lignins are unique biopolymers of plant origin, whose structural organization is multivariate and largely depends on the biological species of a plant. Basic structural and chemical studies on various native lignins increasingly refine our knowledge of the macromolecules structure of this key plant biopolymer, and examining lignins of various taxonomic origins shows to what extent evolution and natural variability have resulted in a complication of the chemical structure of the lignin macromolecules, in particular, through the incorporation of "non-traditional" phenolic monomers. This suggests that lignins contain much more structural units than the three traditional monolignol variants described in lignin chemistry textbooks. As a result, the very definition of lignin continues to be expanded and refined. Currently, there is a dramatic increase in interest in lignin-based materials, mainly due to their diverse beneficial properties such as biodegradability, chemical reactivity, biocompatibility, low toxicity, and a wide range of biological activities. One of the main objectives of this review article was to identify and discuss the mechanisms of the biological action of lignins on living organisms with the view to assessing the biomedical potential and substantiating the possibility of use of lignins as innovative drugs with antioxidant, radiation-protective, and geroprotective effects. Finding new ways to use lignins is necessary to meet the challenge of preservation of health, improvement of quality of life, and extension of life expectancy of people.

Keywords: lignins, structure, botanical species, biomedical properties, pharmacological effects

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1. INTRODUCTION

Natural compounds of plant origin Rare a rich source of diverse compounds for the production of biomedical products. In recent years, plant bioactive substances such as flavonoids, terpenes, lignin-polysaccharide complexes, and lignins have received increasing attention due to the potential benefits for human health they offer [1]. Lignin and cellulose are two most abundant polymers found in plant kingdom. By contrast to cellulose, which is also synthesized by some bacteria (e.g., *Acetobacter aceti*) and by animals such as tunicates (*Tunicata*), lignins of animal origin are unknown. This biopolymer is found in almost all higher land plants, including some bryophytes [2]. Lignins were discovered even in algae, e.g., *Calliarthron cheilosporioide* [3]. At the same time, this biopolymer is not found in lichens and fungi, whose original properties are determined by a combination of specific characteristics inherent in both plants and animals.

The biological functions performed by lignins in plants are very diverse [4]. They impart mechanical strength to the plant tissue as a whole. Lignin provides for rigidity and permeability of cell walls, essential for maintaining cell turgor and for transportation of water, nutrients, and products of metabolism. These biopolymers protect plants against excess exposure to UV radiation and harmful effects of free radicals. Also, lignin increases plant tolerance (adaptive function) to various environmental stresses, e.g., increased radioactive background and extreme temperatures. One of the functions of lignin is antioxidant protection of plants [5]. Another important function is protection against various harmful insects, microorganisms, and other phytopathogens [6].

Thus, lignins create reliable physical, chemical, and biological barriers against adverse impacts in almost all plant elements, including the main stem, vegetative organs (roots, branches, leaves) and reproductive organs (angiosperm fruits, gymnosperm strobili) [7]. In this context, it is logical to enquire as to whether lignin, which protects everything in plants, including reproductive organs, would be able to protect another "child of nature," a man?

For giving a definition to a particular high-molecular-weight compound, it is necessary to elucidate the elemental composition of the polymer, the structural formula of its monomer unit, and the entire set and number of functional groups, as well as its number-average and weight-average molecular weights. In the case of lignin these tasks have not yet been fully accomplished, that is, the problem of elucidation of the chemical structure of lignin is still far from being solved, which complicates employment of the biomedical properties of lignins in practical medicine. The biomedical properties of lignin-based materials depend on their structure, type and number of active functional groups, molecular weight, and botanical (taxonomic) source of lignin, as well as on the method of its extraction from plant tissue [8]. However, universal relationship between the structural organization of lignins and their biological activities have not been fully established and require further study [9]. The results of recent investigations indicate the need to create a database on the structural organization of lignins of different botanical origins [10–12].

The main purpose of this review was to analyze the literature data on the chemical structure of lignin in relation to its biological origin, as well as to assess the biomedical potential of lignins with the view of substantiating the possibility of their use as innovative drugs with antioxidant, radiation-protective, and geroprotective effects.

2. LIGNIN BIOSYNTHESIS

From biochemical perspective, plant lignification consists of two fundamentally different stages: synthesis of primary monomeric lignin precursors (monolignols) and polymerization of these compounds. The first stage begins with formation of glucose during photosynthesis; glucose is converted into shikimic acid, which is the most important intermediate in the so-called shikimate pathway, culminating in the formation of monomeric lignin precursors. The second stage consists in polymerization of the monolignols by a stepwise enzymatic dehydrogenative free-radical polymerization mechanism [13]. The process initiator is the peroxidase-hydrogen peroxide enzymatic complex; enzyme laccase can play a certain role in biosynthesis as well. A fairly broad range of monolignols (precursors) can participate in lignin biosynthesis, among which the main ones are phenylpropane compounds differing in the degree of methoxylation: 4-[(*E*)-3-hydroxyprop-1-enyl]-2-methoxyphenol (coniferyl alcohol), 4-[(*E*)-3-hydroxyprop-1-enyl]-2,6-dimethoxyphenol (sinapyl alcohol), and 4-[(*E*)-3-hydroxyprop-1-enyl]-phenol (*p*-coumaryl alcohol) (Fig. 1).

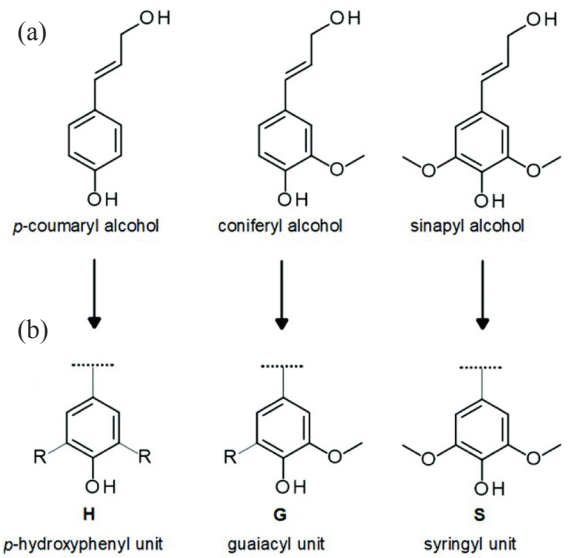


Fig. 1. (a) Monomeric lignin precursors (monolignols) and (b) corresponding general structural units of *p*-coumaryl (H), guaiacyl (G), and syringyl (S) lignin type [14].

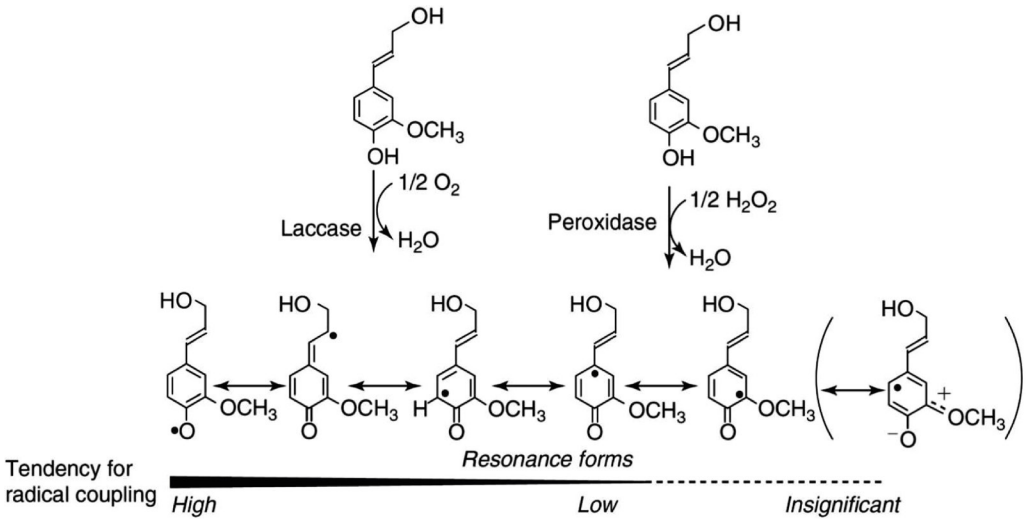


Fig. 2. Enzymatic formation of phenoxy radicals [17].

Enzymatic processes involving peroxidase and laccase lead to the formation of phenoxy radicals, whose reactivities may differ slightly [15, 16] (Fig. 2).

The resultant monomeric phenoxy radicals interact with each other. The above-described mechanism leads to a number of substructures exhibiting a diversity of inter-unit linkages, whereby various microstructures, dilignols, are formed, and this is followed by generation of new, dimeric, phenoxy radicals from dilignol structures. Recombination of mono- and dilignol radicals gives tri- and tetralignols. Subsequent successive dehydrogenation-recombination events lead to the formation of increasingly larger structural moieties and, ultimately, of the lignin macromolecule proper. It should be noted that most researchers hold the opinion that, by contrast to the synthesis of monolignols, the macromolecular lignin assembly, that is, polymerization, is not under genetic control [18–20]. Several types of monolignols are involved in lignin biosynthesis, whereby random coupling of certain precursors results in formation of chemically different macromolecules.

Studying the lignification process *in vivo* is practically impossible, in which situation modeling biosynthesis *in vitro* using both monomeric lignin precursors and

plant tissue cultures constitutes a fruitful approach [21]. Previously, fairly extensive experimental studies were dedicated to polymerization of coniferyl alcohol, sinapyl alcohol, *p*-coumaryl alcohol, 3-methoxy-4-hydroxycinnamic acid (ferulic acid), and 2-hydroxy-3-methoxybenzaldehyde under conditions simulating the biosynthesis of natural lignins in the presence of the peroxidase-hydrogen peroxide enzymatic complex [22–25].

Of particular interest are studies of the dehydrogenative copolymerization of caffeyl alcohol and *p*-coumaryl alcohol, whose polymer product models, to a certain extent, C-lignin, recently discovered in the seed coats of the vanilla orchid [26]. Several computational models, useful for refining the monolignols formation mechanism, were proposed in [27]. The capabilities of analysis with those mathematical models were demonstrated using specific objects, in particular, lignins from black cottonwood *Populus trichocarpa*, alfalfa *Medicago truncatula*, switchgrass *Panicum virgatum,* and the grass *Brachypodium distachyon*. In [28], polymerization of monolignols was presented as a sequence of enzymatic and chemical reactions proceeding in a nonequilibrium open system. The model of the growth of the lignin macromolecule included successive stages of formation of phenoxy radicals of different generation numbers and successive events of recombination of these radicals. The mathematical model of the process appeared as a system of 14 differential equations. The results of the numerical experiments (Runge-Kutta-Fehlberg method) within the framework of the proposed mathematical concept provided theoretical justification to some regularities of the lignification β process and to the lignin structure variability.

3. MOLECULAR STRUCTURE OF LIGNINS

3.1. Structural Units

Lignins from plants of different taxonomic origins differ markedly in the molecular structure and are divided into three main classes: softwood lignins, hardwood lignins, and grass lignins [9]. Wood lignins have been studied fairly thoroughly in response to the needs of the pulp and paper industry. As to lignins of herbaceous plants, increased research attention they have received in recent years is due to a high biomedical potential of these biopolymers, which are distinguished from lignins originated from woody plants by a more complex and multivariant structure [29, 30].

Grass lignins belong to the GSH type, i.e., all the major monomeric lignin precursors are involved in the biosynthesis of these biopolymers. Table 1 presents some of the recently published data on the monomeric composition of lignins isolated from plants of different botanical origins.

As follows from analysis of the data on the three main classes of lignins, discussed above, the most predictable composition is that of softwood lignins, which are constituted by guaiacyl phenylpropane units (PPU) with

Table 1. Distribution of the main structural units in lignins isolated from plants of different botanical origins

Botanical species (plant tissue)	H	G	S	Reference
Triticum (straw)	3	61	36	[29]
Arabidopsis thaliana (xylem)	1.4	73.6	25	[31]
Oryza sativa L. (straw)	5	71	24	[32]
Sesamum (stems)	5	71	24	[33]
Rubus idaeus L. (stems)	0	45.7	54.3	[34]
D. sinicus (stems)	9	35	56	[35]
Helianthus tuberosus (stems)	2.6	44.3	53.1	[36]
Ledum (stems)	0.2	33.5	66.3	[36]
Lavatera (stems)	0.2	31.7	68.1	[36]
Secale sp. (straw)	4.3	46.9	48.7	[36]
Junglas regia (wood)	0.8	29.9	69.3	[36]
Chaenomeles sinensis (wood)	7.6	23	69	[37]
Quercus suber L. (wood)	5	72	23	[38]
Pinus radiate (wood)	0.5	99.5	0	[31]
Picea abies (wood)	0.8	99.2	0	[31]
Pseudostuga menziesii (wood)	0	100	0	[31]

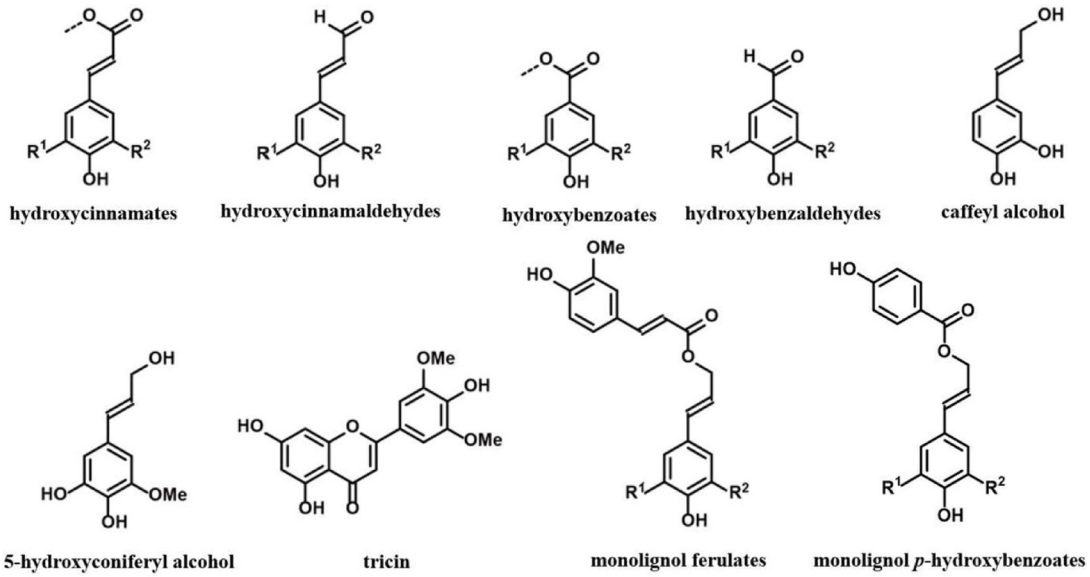


Fig. 3. Non-traditional structural elements of the lignin macromolecules [43].

minor amounts of H-units. At the same time, there are also lignins consisting entirely of G-units, which is the case of, e.g., *Pseudostuga menziesii* [31] and Siberian fir [39]. For lignins of angiosperms, it is very difficult to predict the composition of the macromolecules. Table 2 provides summarized information on the approximate contents of different monomer units in lignins of coniferous, deciduous, and herbaceous plants.

Knowledge about native lignins is still increasing today; in particular, more new structural elements of lignin macromolecules are being discovered. It was shown [9, 42] that, alongside the above-mentioned three canonical monolignols, other phenolic compounds, which are currently considered as potentially possible monomeric units, may also participate in plant lignification. These include caffeyl alcohol, 5-hydroxyconiferyl alcohol, hydroxybenzaldehyde, and some other aromatic compounds (Fig. 3).

Typical monolignol ester conjugates with ferulates and *p*-hydroxybenzoates can also behave as monomeric units and participate in macromolecular lignin assembly [43]. In this context, of great interest are studies [44–46] reporting the presence of so-called catechyl lignin (C-lignin), consisting of caffeyl alcohol units. The main feature of this lignin is that this polymer, by contrast to classical lignins, is built from structural units of only one type, i.e., is a homopolymer. As indicated in [45, 47], unlike conventional GS lignin, C-lignin is a homogeneous and linear biopolymer, consisting of simplex caffeyl alcohol subunits with benzodioxane as the dominant linkages, which exhibits enhanced resistance under harsh (acid) and mild (enzymatic) treatments. When coexisting with G/S-lignin, e.g., in seed coats of Orchidaceae, Cactaceae, Euphorbiaceous, and Cleomaceae plants, naturally occurring C-lignin can be separated, owing to its higher acid resistance. The homogeneity of C-lignin allows its depolymerization into 1,2-dihydroxybenzene suitable for downstream processing with nearly 100% theoretical maximum monomer yield. As concluded in [45], the homogeneity and linearity of C-lignin render it an "ideal lignin" archetype for a high-cost commercial product.

Table 2. Distribution (%) of the structural units of the different types in lignins for plants of different biological origins [40, 41]

Structural unit	Softwood	Hardwood	Grass
S	0–1	50–75	25–50
G	90–95	25–50	25–50
H	0.5–3.4	Traces	10–25

Over the past few years, it has also been established that some phenolic compounds from beyond the canonical monolignol biosynthetic pathway may behave as true lignin monomers. They participate in various coupling and cross-coupling reactions with classical monolignols and are integrally incorporated into typical lignins [48]. Among these compounds is tricin (4′,5,7-trihydroxy-3′,5′-dimethoxyflavone) (Fig. 3), which was found incorporated into lignins of rice straw, sorghum, and some other plants [49–51].

Analysis of the most recent data [52, 53] shows that tricin is quite widespread in lignins of monocotyledonous plants, and especially in cereals of the Poaceae family. Various amounts of tricin were found in stems, aerial roots, and seeds of *Vanilla planifolia* [26, 49]. Information about the presence of tricin-containing lignin in some plants is of no small importance. Firstly, it shows that tricin can act as a monomeric unit, and secondly, it is indicative of a high potential value of lignin as a feedstock for production of new preparations, e.g., nutraceuticals, with various biological activities.

Another new structure, piceatannol [54, 55], was discovered in the palm fruits endocarp tissues. Those studies showed that piceatannol and some other hydroxystilbenes, in particular, resveratrol and isorhapontigenin, are incorporated into the lignin macromolecule via cross-coupling with typical monolignols. It became obvious that continued research into lignins of various botanical origins would lead to the discovery of new types and kinds of this biopolymer. This prediction was confirmed by research [56] which demonstrated a unique structure of lignin polymer of papyrus (*Cyperus papyrus* L.) and described it as the only lignin known to date that integrates such a diversity of phenolic compounds from different classes of flavonoids, including naringenin, tricin, and dihydrotricin. This information might open up new opportunities for using lignins as biomedical products with specific properties and for enhanced value.

To theoretically substantiate the possibility of the existence of nontypical monomeric lignin precursors, computational approaches were applied [57]. Specifically, density functional theory calculations of the thermodynamics of the reactions of bond dissociation, coupling, and rearomatization of lignin monomers such as tricin, caffeyl alcohol, 5-hydroxyconiferyl alcohol, and piceatannol were carried out. The theoretical possibility of incorporation of the above-mentioned structures into the lignin macromolecule was demonstrated. Thus, computational methods help unravel important molecular underpinnings of biosynthesis involving a broad range of phenolic compounds. The discovery of non-traditional phenolic precursors arising from other biosynthetic pathways than the canonical monolignol biosynthetic pathway expands the traditional definition of lignin and reveals that any phenolic compound that is delivered to the cell wall may be oxidized and incorporated into the lignin polymer during lignification, subject exclusively to its chemical compatibility [58].

3.2. Inter-Unit Linkages

A remarkable feature of lignins as polymers is the diversity of chemical bonds linking their structural units; no other biopolymers with similar specificity is found in nature. The most common inter-unit linkage in the lignin polymer is β-*O*-4 bond, forming the structures of alkyl-aryl ethers. Also, the presence of bonds (and corresponding structures) such as β-5 (phenylcoumarans), β-β (resinols), 5-5 (dibenzodioxocins), 5-*O*-4 (biphenyl ethers), and β-1 (spirodienones) was revealed in lignin (Fig. 4).

The composition and content of these structures can vary significantly depending on the species, growing conditions, and maturity stage of a plant, as well as on the type of plant cells [60, 61]. For example, study of the structure of lignin from different tissues of the plant *Vanilla planifolia* (aerial roots, nodes, internodes, and seeds) using advanced nuclear magnetic resonance methods [49] showed that the lignin from the nodes and internodes displayed traditional S/G type lignin type with 65 and 73% β-*O*-4′ bonds in the nodes and internodes lignin, respectively. The lignin of the aerial roots of this plant contained 96% β-alkyl-aryl ether linkages. Table 3 presents some recently published data on the distribution of the different inter-unit linkages in lignins of different taxonomic origins.

Table 4 provides summarized information on the distribution of the different types of inter-unit linkages and dilignols in lignins of the different biological origins. It should be noted that the currently available knowledge does not allow any realistic predictions of the chemical structure of the previously unstudied lignins. Possibly, predictions for lignins of coniferous species may be more or less close to reality.

It is also necessary to mention the influence of exogenous factors [67] on the structural organization of lignins, in particular, on the distribution of the different

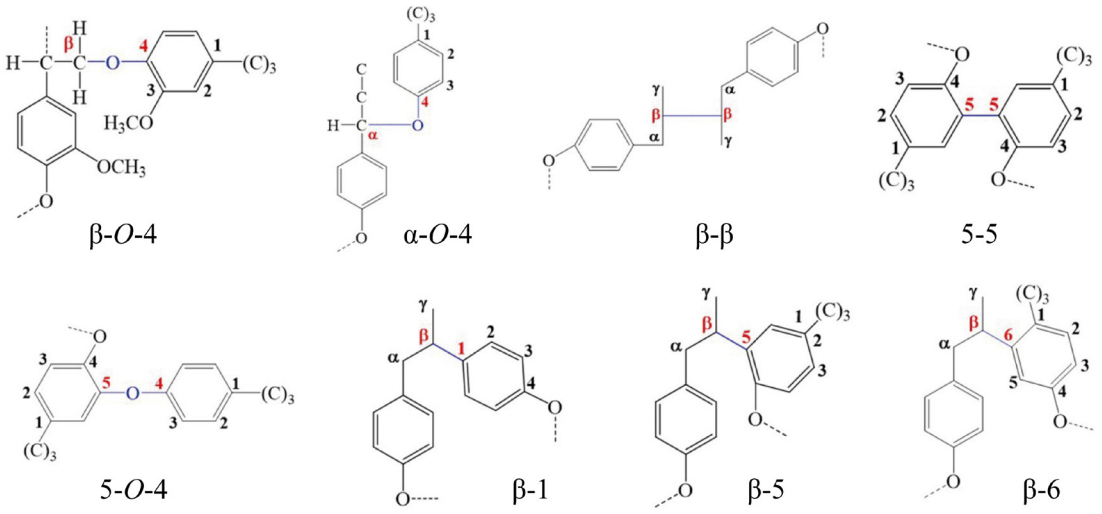


Fig. 4. Principal inter-unit linkages in lignin [59].

types of inter-unit linkages and on the number of the corresponding structural units in their macromolecules. At sufficiently high levels of negative exogenous effects on plants, physiological processes can lead to disruption of the conventional lignin biosynthesis pathway and to certain changes in the chemical structure of the lignin macromolecule [4, 14, 68]. In this regard, the lignin polymer synthesized by plants in response to stresses is called stress lignin (or *defense lignin*) to distinguish it from typical lignin (*developmental lignin*) [69]. Recent findings on the oxidative stress caused in plants by ozone exposure suggest that the structure of the lignin polymer largely depends both on the intensity and type of the stress and on the plant species [70]. New data [60, 71, 72] indicate that some abiotic stresses, such as ultraviolet radiation, low temperatures, and gravitational stress, cause change not only in the degree of lignification of the plant tissues but also in the lignin structure of woody plants. According

Table 3. Distribution (number/100 PPU) of the principal inter-unit linkages in lignins of different botanical origins

Botanical species (plant tissue)	β-*O*-4	β-5	β-β	β-1	Reference
Rubus idaeus L (stems)	54.8	3.9	11.3	0.92	[34]
Triticum (straw)	39.8	7.8	0.27	–	[29]
Oryza sativa L. (straw)	78	12	4	1	[32]
D. sinicus (stems)	79.5	5.6	7.2	4	[35]
Chaenomeles sinensis (wood)	74.1	6.1	19.8	–	[37]
Spruce (wood)	38	10	2	1	[62]
Populus × *euramericana cv.* (wood)	51.2	8.26	7.85	4.0	[63]

Table 4. Distribution (%) of the different types of inter-unit linkages in lignins isolated from herbaceous, coniferous, and deciduous plants [64–66]

Linkage type	Dimer structure	Grasses	Hardwood	Softwood
β-*O*-4	β-alkyl-aryl ether	69–94	50–65	45–50
a-*O*-4	a-alkyl-aryl ether	11–14	1–8	5–8
β-5	Phenylcoumarans	5–11	3–11	9–12
5-5	Biphenyls and dibenzodioxocins	n.i.	1–5	5–25
5-*O*-4	Diaryl ether	n.i.	6–7	3.5–8
β-1	Spirodienones	0–2	1–7	1–10
β-β	Resinols	1–15	3–12	2–6

to [73], water deficiency causes osmotic stress in plants exposed to drought. Simulation of drought in experiments with *Eucalyptus urograndis* revealed an altered chemical structure of lignin. In particular, it was demonstrated that the stress lignin macromolecules differed from those of typical lignins by a higher content of S and a reduced content of G monolignol units. Other studies [60, 74] showed that stress lignins differed from typical lignin by an increased content of H-type monolignols and of C–C inter-unit bonds. It was concluded that high concentrations of H-type monolignols in that case contributed to the formation of more condensed lignin with a higher number of β-5 and β-β inter-unit bonds than in developmental lignin. Certainly, a change in the chemical structure of one of the main plant components cannot but affect the utilitarian, practically useful, in particular, biomedical properties of plants, which makes relevant studies of the chemical structure of not only typical but also stress lignins.

3.3. Functional Groups

Unlike plant polysaccharides, lignin is a multifunctional polymer, which constitutes a very important advantage of lignins over other biopolymers in terms of the biomedical potential offered by drugs on their basis. Lignin contains almost all known oxygen-containing functional groups: methoxyl, phenolic, carbonyl, and aldehyde, as well as aliphatic hydroxyls and carboxyls. All the known laws governing the organic chemistry reactions are valid for the functional groups of lignin, which fact will certainly be useful for the search of new ways to use lignin, in particular, in biomedicine [75].

Very high abundance of OCH_3 groups has always been considered to be a characteristic feature of lignins, with methoxyl content being a criterion of the purity and "nativity" of isolated preparations. These groups are located in the aromatic ring at positions 3 in guaiacyl PPU and at positions 3 and 5 in syringyl structural units. However, as mentioned above, some plants can synthesize C-lignins, which do not contain methoxy groups. This brings into challenge another axiom, specifically that stating that lignin is always a methoxylated polymer. Table 5 presents the data on the methoxyl content in selected lignin samples. As shown by analysis of literature data, the weight percent of methoxyls in guaiacyl lignins does not exceed 17, and in guaiacyl-syringyl lignins it reaches 22. In non-woody plants, the content of methoxy groups varies more widely than in woody plants [75]. However, their quantity depends not only on the origin of lignin but also on its localization in various elements and parts of the plant. It is not unlikely that the degree of methoxylation is essential from the biomedical viewpoint, since lignins with a higher content of OCH_3 groups in some cases show a higher antimicrobial activity, which was attributed in [76] to certain antiradical properties of methoxy groups.

Important functionalities of lignin are hydroxyl and carboxyl groups, since they determine hydrophilicity–hydrophobicity of the macromolecules and significantly influence the lignin behavior during hydrolysis processes, e.g., under gastrointestinal tract conditions. High Ar-OH and low Alk-OH content introduce resistance to hydrolysis of lignin [77]. Carboxyl groups (–COOH) influence the physicochemical properties of lignin due to creating electrostatic charge and enhancing hydrogen bonding. Ar-OH and COOH groups play an important role in enzymatic hydrolysis reactions, but their correlation with these processes becomes complicated due to the specificity of lignins of various botanical origins. Table 5 lists the contents of the major functional groups in lignins of deciduous and coniferous species.

All native lignins contain phenolic OH groups whose content significantly depends on the origin of lignin and, in the case of isolated preparations, on the method of isolation from the plant tissue. Phenolic groups play a critical role in many biochemical processes [78, 79] due

Table 5. Functional group composition (number/100 PPU) in selected lignins [75]

Functional groups	Hardwood lignin	Softwood lignin
Methoxy groups	130–146	90–96
Phenolic groups	9–20	20–28
Benzyl groups	–	16
Aliphatic hydroxyls	–	120
Carbonyl groups	3–7	20
Carboxyl groups	11–13	–

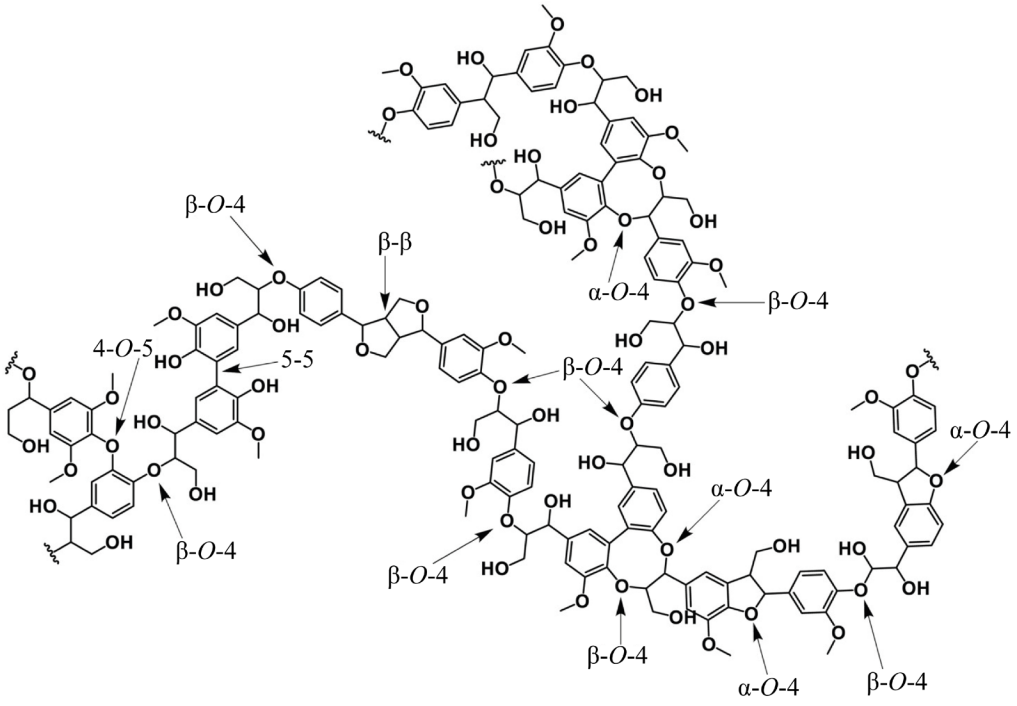


Fig. 5. Typical lignin structure [5].

to their high reactivity. According to [8], some types of lignins can act as natural antioxidants by suppressing a number of oxidative processes in mammals. The antioxidant activity of lignins may well be mainly associated with the effect of phenolic structures on toxic free radicals [65].

Thus, the recently published data significantly expand and refine the current understanding of the chemical structure of lignins. Progress in the field of lignin chemistry was largely driven by the use of new advanced research methods, including ^{13}C NMR, ESR, and FT-IR spectroscopy and pyrolysis-gas chromatography/mass spectrometry, as well as by the expansion of the range of plant objects and, to a certain extent, by the abandonment of some outdated dogmas and theories.

Yet there has been no fundamental change in understanding the chemical structure of lignin macromolecules, and most researchers hold the conventional views on this subject, as demonstrated by its recently published structural model (Fig. 5), based on the established views on the structure of this biopolymer. In this regard, it becomes obvious that studies of the structural organization of lignins, including the features of its molecular structure, must be continued in view of remarkable biological activity possessed by lignins, which provides for a real possibility of using them in pharmaceutical biotechnology and biomedicine.

4. BIOMEDICAL PROPERTIES OF LIGNINS

In recent years, lignin has attracted increasing attention from researchers that have shown interest in the prospects of its diverse biomedical and pharmaceutical applications, ranging from enteral nutrition and treatment of obesity to tissue engineering and cosmetics. Literature analysis reveals a substantial increase in the number of publications dedicated to biomedical uses of lignin-based preparations. At the same time, only a fairly limited number of review articles on this topic is available, in spite of constantly growing demand for summarization of the results of studies on biomedical uses of lignins. Below, we will focus on publications demonstrating various pharmacological effects, including geroprotective, antioxidant, and radiation-protective activities of agents based on native and technical lignins.

4.1. Pharmacological Effects of Lignins

An analysis of the current state of research on lignin and its derivatives has revealed growth of interest in this scientific area due to the multifaceted effects produced by these compounds on human organism, as well as to

the lack or insignificant manifestation of side effects from their application. It has now been established that phenolic and polyphenolic compounds produce beneficial effects on human health, including protection against oxidative stress, cardiovascular diseases, diabetes, asthma, neurodegenerative diseases, and aging [80–82]. Specifically, a decrease of oxidative and inflammatory damage to the kidney in streptozotocin-induced diabetic rats due to lignin-derived lignophenols was reported [83]. Lignophenols improve lipid metabolism, which may be useful in treating obesity [81]. Testing sulfated low-molecular-weight lignins *in vitro* for their efficiency against emphysema revealed anti-inflammatory effect [84]. Anticoagulant and antiplatelet action of lignin derivatives was reported in [85]. Particularly interesting effects of lignin include an increase in density of some intestinal immunocompetent cells without exerting effects on those in the peripheral blood [86], as well as improvement of cardiovascular parameters, in particular, endothelial function, arterial stiffness, blood pressure, cholesterol, and glycemic control [87]. Lignins may be involved in accelerating elimination of cholesterol and other fat-soluble substances from the body, which can serve as a physiological mechanism for lowering blood cholesterol in vegetarian diets and preventing cardiovascular diseases [88].

A number of studies were dedicated to the antitumor effect of lignin derivatives. Specifically, plant polyphenolic compounds have been shown to inhibit chemically induced cancer in the lungs, esophagus, intestines, liver, and skin of laboratory animals [89–91]. The anticarcinogenic activity of plant-derived polyphenols is manifested in the inhibitory effects on the proliferation of cancer cells, tumor growth, metastasis, inflammation, and induction of apoptosis [92–95].

Lignins are also considered as potential drug [81, 82, 92, 96, 97] and gene [98] delivery vehicles and are also used in tissue engineering [99] as a tool to restore, replace, and improve the function of a specific tissue or organ [82, 95, 99, 100] and in wound healing [101].

Many studies have shown the presence of antimicrobial activity in lignins [82, 95, 97, 102–108]. Also, lignosulfonic acid and lignin-carbohydrate complexes have been studied for their antiviral capacity [109, 110]. In particular, lignosulfonic acid exhibits a broad-spectrum anti-HIV activity and has potential for microbicidal applications [111]. Antiviral activity has been established for lignin-carbohydrate-protein complexes from *Pimpinella anisum* seed extract [112].

Due to its antimicrobial [102], as well as antifungal [95] properties, lignin can be considered a

potential source of biopesticides. For example, a complex consisting of lignin nanoparticles and polylactic acid is capable of inhibiting bacterial growth [113]. Polyphenolic compounds of lignin cause cell membrane damage and lysis, i.e., destruction and dissolution of bacteria with subsequent release of cellular contents [114]. It was shown [115] that lignins can be applied to stabilize food and feedstuff owing to their antifungal, antiparasitic, and antioxidant properties. Hydrolysis lignin has long been used as an enterosorbent [95, 108, 116–118]. Lignins of food fibers have high affinity to sex steroid hormones, which provides their ability to interrupt enterohepatic circulation of hormones and to accelerate their removing from organism. High sorption capacity of lignins (by contrast to hemicelluloses and cellulose) for the steroid hormones progesterone and estradiol was demonstrated [119]. Specifically, in experiments performed *in vitro* and *in vivo*, 90% sorption of progesterone on lignin was achieved. A hypothesis was advanced about the key role of native lignins in maintaining the balance of sex hormones in the mammalian body. Studies have been carried out on the adsorption of sex steroid hormones on samples of lignins and lignin-pectin globules [88, 119]. Those data indicate the possibility of forming stable supramolecular complexes from citrus pectin and natural lignins for the production of pharmacological preparations to be employed in targeted drug delivery.

4.2. Antioxidant Activity of Lignins

A characteristic feature of lignin macromolecules is the presence of several types of phenolic groups with different *p*K values, as well as of phenoxy radicals, which makes lignins structurally similar to low-molecular-weight antioxidants of plant origin [78, 120]. There is a growing body of literature indicating that lignin is an effective free radical scavenger able to reduce oxygen radicals and to stabilize oxidation reactions [59, 81, 82, 95, 102, 106, 121–123]. In the gastrointestinal tract of mammals, lignins are partially converted to low-molecular-weight biphenolic compounds, lignans [88]. Two lignans, enterodiol and enterolactone, found in

significant quantities in the blood, milk, and urine of vegetarians and herbivores, have antioxidant properties. Most antioxidant effects of lignins are considered to be derived from the scavenging action of their phenolic structures on free radicals. Therefore, the reducing power of a compound can serve as a significant indicator of its potential antioxidant activity [124]. It was reported [125] that lignin converted into a nanoparticle has increased antioxidant activity due to the large number of phenolic groups on the nanoparticle surface.

Natural antioxidants typically regulate the degree of influence of non-enzymatic free radical-mediated oxidation on most biochemical processes in the body, thereby creating optimal conditions for metabolism and ensuring the normal functioning of body cells and tissues [126]. The authors of [127] were among the first to draw attention to the antioxidant properties of lignins and to propose a method for producing water-soluble lignins. Using this method, preparations, practically not inferior in antioxidant activity to synthetic antioxidants widely used in medical practice, were obtained from cereal straw. A conclusion that lignins make a significant contribution to the overall antioxidant activity of plant-based foods was made in a number of studies [126, 128]. It was also noted that the ability of lignin to scavenge oxygen radicals is an additional advantage they offer for drug delivery in the treatment of a number of diseases that generate reactive oxygen species [129]. In that study, the effect of Olipifat agent on the free radical process was observed as part of *in vivo* experiments. A particularly clear antioxidant and antiradical effect of Olipifat was observed in experiments on rats inoculated with Walker carcinosarcoma. Increased production of free radicals and a reduced antioxidant defense level characteristic of tumor-bearing rats were significantly normalized from administering Olipifat. Specifically, the level of production of reactive oxygen species in the blood and tumor tissue decreased, and the overall antioxidant and antiradical activity of the studied tissues increased significantly.

4.3. Radiation-Protective Properties of Lignins

Radiation is a constant companion of life in the modern world. Exposure of a living organism to ionizing radiation can induce a chain of reversible and irreversible effects, in particular, radiolysis of water with the formation of reactive oxygen species [130], which cause damage to various organs and systems of the mammalian body. Development of a single concept of radiation protection of humans and the environment is the focus of serious attention by various international organizations (IAEA, UNSCEAR, UN, ICRP, EU, WHO, etc.), addressing not only environmental, but also legal, ethical, and social issues related to radiation safety problems. Currently, active screening of substances that mitigate or neutralize the harmful effects of radiation is underway. The difficulty of finding substances suitable as radioprotectors consists in that the candidates must simultaneously satisfy a number of requirements: high efficiency, lack of toxicity, convenient administration route, low manufacturing costs, storage stability, long duration of protective action, and the ability to alleviate effects of various types of ionizing radiation [131]. Studies conducted in a number of countries of the world (India, China, etc.) consider natural antioxidants, plant products, as radioprotectors, that are well tolerated, exhibit high efficacy, and are available and obtainable in large quantities [132]. Much promise in this regard is offered by lignin-based preparations. Recently, experimental studies in which lignin and its derivatives were proposed to be used as radioprotective agents were reported [133]. Specifically, in mice exposed to gamma irradiation, a novel lignin-derived polyphenolic composition with ammonium molybdate demonstrated radioprotective potential mediated through hematopoietic and intestinal stem progenitor cells [134, 135]. The radiation-protective effect of this compound is associated with the antioxidant activity of its polyphenolic ligand. Accordingly, one of the most likely mechanisms for the radiation-protective action of various lignin derivatives [78, 120] is underlain by the ability of these compounds to reduce the level of reactive free radicals [136] and inhibit gamma radiation-induced lipid peroxidation [137]. Another mechanism of action of plant complexes of polyphenolic origin is to neutralize the negative impact of ionizing radiation and chemical toxicants via accelerating DNA repair or induction of apoptosis in cells with "critical" genetic changes. These effects were observed in a wide group of polyphenolic compounds, including lignin [138]. Various natural substances have been shown to effectively mitigate DNA damage induced by various genotoxic agents. The potential for protection against strand breaks was demonstrated for various polyphenolic compounds of plant origin in a comet assay used in animal experiments [139]. Also, a number of studies revealed the adsorptive action of lignin derivatives on DNA-damaging agents [140].

Regarding effects produced by radiation exposures on humans and animals, use of radiation therapy to treat malignant and benign tumors should be mentioned alongside negative effects from high radiation doses. Currently, there is an acute problem of finding new ways to reduce the side effects of antitumor treatment. Research is being conducted into the possibility of using low-toxic agents of biological origin as potential anticancer agents. In this respect, plant polyphenolic compounds are being evaluated in both experimental and clinical studies. Compounds whose chemopreventive and chemotherapeutic potential was studied most extensively include saponins, carotenoids, curcumin, quercetin, and resveratrol [141–144], while lignins remain a group of polyphenolic compounds that are the least studied in this respect [133]. A few studies were reported in which lignin derivatives were proposed as accompanying therapy for cancer patients to accelerate the natural processes of damage repair. Also, it has been established that lignin bear chromophoric groups that can absorb light in the ultraviolet region [63, 81, 82, 95, 102, 103, 108, 122].

4.4. Geroprotective Properties of Lignins

Radiation-protective and anti-aging properties are often combined, which is due to the common mechanisms of action: antioxidant activity, enhancement of resistance to stress impacts at the cellular level, acceleration of DNA repair, and prevention of chronic diseases and carcinogenesis [146]. In this connection, it was pointed out that, on the one hand, against the background of stopping the cell cycle and blockade of apoptosis the time required to repair the defects of a DNA increases, and, on the other hand, activation of apoptosis of altered cells can be seen as one of the mechanisms to delay aging processes and prevention of isolated effects of exposure to ionizing radiation. Two main groups of radiation-protective drugs have been identified: (1) those possessing anti-apoptotic properties and helping increase survival rates in the early post-irradiation period after high-dose exposure and (2) those promoting the elimination of damaged cells (senolytics), most effective in the case of long-term exposure to low-dose radiation or fractionated radiation. Geroprotective activity has been described for drugs of the second group, including melatonin, metformin, rapamycin, and some natural polyphenolic compounds. Lignins isolated from medical grassy plants *Rhodiola rosea* L. and *Serratula coronata* L. were shown to possess pronounced geroprotective properties, causing an increase in the life span of *Drosophila melanogaster* males [147, 148]. Geroprotective properties of lignins were revealed in experiments using small mammals [88, 149].

Thus, as shown by analysis of recent publications, the pharmacological effectiveness of lignin-based drugs can be considered an established fact. Table 6 provides

Table 6. Biological effectiveness of lignins

Biological effect	Reference
Protection against oxidative stress in cardiovascular diseases, diabetes, asthma, neurodegenerative diseases and aging, blood cholesterol lowering	[80–82]
Obesity treatment	[81]
Anticoagulant and antiplatelet activity	[85]
Anticarcinogenic activity	[89–95]
Potential nanosystems for drug and gene delivery	[81, 82, 97, 100]
Tissue engineering	[82, 95, 99, 100, 101]
Antimicrobial action	[8, 82, 95, 97, 98, 102, 103, 105–107]
Antiviral action	[109–112]
Antimicrobial and antifungal properties: biopesticides; stabilization of food and feedstuff	[95, 102, 113]
High sorption capacity	[95, 108, 110, 116, 117]
Antioxidant activity	[8, 81, 95, 102, 106, 121–123]
Radiation-protective properties	[133–137]
Acceleration of DNA repair or induction of apoptosis	[138, 140]
UV radiation absorption	[8, 81, 82, 95, 102, 103, 108, 122, 145]
Geroprotective properties	[88, 146–149]

summarized information on the biological effectiveness of lignins.

5. CONCLUSIONS

Thus, analysis of the recently published scientific data significantly deepens and refines current knowledge of the chemical structure of lignins. Progress in the field of lignin chemistry was largely driven by the use of advanced research methods, including 2D NMR, ^{13}C NMR, ESR spectroscopy, mass spectrometry, and pyrolysis-gas chromatography/mass spectrometry, as well as by the expansion of the range of plant objects and the emergence of new concepts of biosynthesis and structural organization of this biopolymer. Revealing "non-traditional" phenolic precursors, resulting from biosynthetic pathways other than the canonical biosynthetic pathway of the well-known monolignols, expands the traditional definition of lignin and shows that a number of new phenolic compounds can be incorporated into the macromolecule during lignification of a plant organism. The results of recent studies confirm the view of lignin as a kind of polymer having a chemical structure as unique as fingerprint in plants of various biological origins. The concept of the biomedical value of lignins provides a basis for the creation of a new generation of highly demanded therapeutic and prophylactic preparations. This review highlights the biomedical potential of lignins in the development of innovative biologically active drugs with antioxidant, radiation-protective, and geroprotective properties, as well as of preventives to reduce the risk of cancer. It should be noted, however, that one of the problems in obtaining lignin-based drugs is that of standardization, because the chemical composition and structure of the macromolecules of this biopolymer depend on a number of factors, including, alongside taxonomic origin, various endo- and exogenous (stress) factors, as well as methods of its isolation. Therefore, further research on lignins from various plants is needed to identify reliable relationships among the botanical origin of plants, the structure of the macromolecules, and the biological effects of lignins.

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This article does not contain any studies involving patients or animals as test objects.

Informed consent was not required for this article.

CONFLICT OF INTEREST

No conflicts of interest was declared by the authors.

AUTHOR CONTRIBUTION

All authors made an equal contribution to the writing of this article.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

1. Zhang, C., Liu, Y., Liu, X., Chen, X., and Chen, R., *Molecules*, 2023, vol. 28, Article ID: 365.
2. Karmanov, A.P., Kocheva, L.S., and Karmanova, Yu.A., *Khim. Rast. Syr'ya*, 2015, pp. 109–114.
3. Xue, J.Y., Hind, K.R., Lemay, M.A., Mcminigal, A., Jourdain, E., Chan, C.X., and Martone, P.T., *PLoS One*, 2022, vol. 17, Article ID: e0266892. https://doi.org/10.1371/journal.pone.0266892
4. Han, X., Zhao, Y., Chen, Y., Xu, J., and Jiang, C., *For. Res.*, 2022, vol. 2, Article ID: 9. https://doi.org/10.48130/FR-2022-0009
5. Lu, X., Gu, X., and Shi, Y., *Int. J. Biol. Macromol.*, 2022, vol. 210, pp. 716–741. https://doi.org/10.1016/j.ijbiomac.2022.04.228
6. Domínguez-Robles, J., Cárcamo-Martínez, Á., Stewart, S.A., Donnelly, R.F., Larrañeta, E., and Borrega, M., *Sustain. Chem. Pharm.*, 2020, vol. 18, Article ID: 100320.
7. Liu, Q., Luo, L., and Zheng, L., *Int. J. Mol. Sci.*, 2018, vol. 19, Article ID: 335. https://doi.org/10.3390/ijms19020335
8. Sugiarto, S., Leow, Y., Tan, C.L., Wang, G., and Kai, D., *Bioact. Mater.*, 2022, vol. 8, pp. 71–94.
9. Ralph, J., Lapierre, C., and Boerjan, W., *Curr. Opin. Biotechnol.*, 2019, vol. 56, pp. 240–249. https://doi.org/10.1016/j.copbio.2019.02.019
10. Rencoret, J., Kim, H., Evaristo, A.B., Gutiérrez, A., Ralph, J., and Del Río, J.C., *J. Agric. Food Chem.*, 2018, vol. 66, pp. 138–153.

11. Reyt, G., Ramakrishna, P., Salas-González, I., Fujita, S., Love, A., Tiemessen, D., Lapierre, C., Morree, K., Calvo-Polanco, M., Flis, P., Geldner, N., Boursiac, Ya., Boerjan, W., George, M.W., Castrillo, G., and Salt, D.E., *Nat. Comm.*, 2021, vol. 12, pp. 1–15. https://doi.org/10.1038/s41467-021-22550-0
12. Mottiar, Y., Smith, R.A., Karlen, S.D., Ralph, J., and Mansfield, S.D., *New Phytol.*, 2022, vol. 237, pp. 251–264. https://doi.org/10.1111/nph.18518
13. Giummarella, N., Pu, Y., Ragauskas, A. J., and Lawoko, M., *Green Chem.*, vol. 21, pp. 1573–1595.
14. Österberg, M., Sipponen, M.H., Mattos, B.D., and Rojas, O.J., *Green Chem.*, 2020, vol. 22, pp. 2712–2733.
15. De Meester, B., Vanholme, R., Mota, T., and Boerjan, W., *Plant Comm.*, 2022, vol. 3, Article ID: 100465. https://doi.org/10.1016/j.xplc.2022.100465
16. Dixon, R.A. and Barros, J., *Open Biol.*, 2019, vol. 9, Article ID: 190215.
17. Gellerstedt, G., Henriksson, G., Belgacem, M.N., and Gandini, A., *Monomers, Polymers, and Composites from Renewable Resources*, Belgacem, M.N. and Gandini, A., Eds., Amsterdam, Elsevier, 2008, pp. 201–224.
18. Ralph, J., Brunow, G., Harris, P.J., Dixon, R.A., Schatz, P.F., and Boerjan, W., In: *Recent Advances in Polyphenol Research*, Daayf, F. and Lattanzio, V., Eds., Hoboken, Wiley-Blackwell, 2008, vol. 1, pp. 36–66.
19. Zhao, Q., *Trends Plant Sci.*, 2016, vol. 21, pp. 713–721.
20. Mottiar, Y., Vanholme, R., Boerjan, W., Ralph, J., and Mansfield, S.D., *Curr. Opin. Biotechnol.*, 2016, vol. 37, pp. 190–200.
21. Kärkönen, A. and Koutaniemi, S., *J. Integr. Plant Biol.*, 2010, vol. 52, pp. 176–185.-
22. Shao-Ying, G., Mlynár, J., and Sarkanen, S., *Phytochemistry*, 1997, vol. 45, pp. 911–918. https://doi.org/10.1016/S0031-9422(97)00077-0.
23. Karmanov, A.P., Matveev, D.V., and Monakov, Yu.B., *Dokl. Chem.*, 2001, vol. 380, pp. 280–283.
24. Li, Q., Koda, K., Yoshinaga, A., Takabe, K., Shimomura, M., and Hirai, Y., *J. Agric. Food Chem.*, 2015, vol. 63, pp. 4613–4620.
25. Lyu, Y., Suzuki, S., Nagano, H., Shigetomi, K., Tamai, Y., Tsutsumi, Y., and Uraki, Y., *Carbohydr. Polym.*, 2023, vol. 301, Article ID: 120305.
26. Chen, F., Tobimatsu, Y., Havkin-Frenkel, D., Dixon, R.A., and Ralph, J., *Proc. Natl. Acad. Sci.*, 2012, vol. 109, pp. 1772–1777.
27. Faraji, M., Fonseca, L.L., Escamilla-Treviño, L., Barros-Rios, J., Engle, N., and Yang, Z.K., *Biotechnol. Biofuels*, 2018, vol. 11, pp. 1–17.
28. Karmanov, A.P. and Poleshchikov, S.M., *Khim, Rast. Syr'ya*, 2019, no. 1, pp. 63–72.
29. Zhang, L., Larsson, A., Moldin, A., and Edlund, U., *Ind. Crops Prod.*, 2022, vol. 187, Article ID: 115432.
30. Karmanov, A.P., Kanarsky, A.V., Kocheva, L.S., Belyy, V.A., Semenov, E.I., Rachkova, N.G., Bogdanovich, N.I., and Pokryshkin, S.A., *Polymer*, 2021, vol. 220, Article ID: 123571. https://doi.org/10.1016/j.polymer.2021.123571
31. Pesquet, E., Wagner, A., and Grabber, J.H., *Curr. Opin. Biotechnol.*, 2019, vol. 56, pp. 215–222. https://doi.org/10.1016/j.copbio.2019.02001
32. Rosado, M.J., Rencoret, J., Marques, G., Gutiérrez, A., and Del Río, J.C., *Front. Plant Sci.*, 2021, vol. 12, Article ID: 640475. https://doi.org/10.3389/fpls.2021.640475
33. Qin, Z., Yang, Q.L., Cheng, X.C., Liu, H.M., and Wang, X.D., *Ind. Crops Prod.*, 2021, vol. 169, Article ID: 113677. https://doi.org/10.1016/j.indorop.2021.113677
34. Popova, Y.A., Shestakov, S.L., Belesov, A.V., Pikovskoi, I.I., and Kozhevnikov, A.Y., *Int. J. Biol. Macromol*., 2020, vol. 164, pp. 3814–3822. https://doi.org/10.1016/j.ijbiomac.2020.08.240
35. Shi, Z., Xu, G., Deng, J., Dong, M., Murugadoss, V., and Liu, C., *Green Chem. Lett. Rev.*, 2019, vol. 12, pp. 235–243. https://doi.org/10.1080/17518253.2019.1627428
36. Karmanov, A.P., Kanarsky, A.V., Kocheva, L.S., Semenov, E.I., and Belyy, V.A., *React. Funct. Polym.*, 2021, vol. 167, Article ID: 105033. https://doi.org/10.1016/j.reactfunctpolym.2021.105033
37. Wang, W.Y., Qin, Z., Liu, H.M., Wang, X.D., Gao, J.H., and Qin, G.Y., *Molecules*, 2021, vol. 26, Article ID: 398. https://doi.org/10.3390/molecules26020398
38. Branco, D.G., Santiago, C., Lourenco, A., Cabrita, L., and Evtuguin, D.V., *J. Agric. Food Chem*., 2021, vol. 69, pp. 8555–8564. https://doi.org/10.1021/acs.jafc.1c01961
39. Karmanov, A.P., Kocheva, L.S., Belyaev, V.Y., Brovko, O.S., and Ovodov, Y.S., *Izv. Vyssh. Uchebn. Zaved., Lesn. Zh.*, 2013, no. 1, pp. 137–145.
40. Wang, J. and Wang, S., *J. Cleaner Prod.*, 2019, vol. 227, pp. 1002–1022.

41. Yuan, J.M., Li, H., Xiao, L.P., Wang, T.P., Ren, W.F., Lu, Q., and Sun, R.C., *Fuel*, 2022, vol. 319, Article ID: 123758.
42. Smith, R.A., Lu, F., Muro-Villanueva, F., Cusumano, J.C., Chapple, C., and Ralph, J., *Plant Cell Physiol.*, 2022, vol. 63, pp. 744–754. https://doi.org/10.1093/pcp/pcac051
43. Sun, R.C., *ChemSusChem*., 2020, vol. 13, pp. 4385–4393.
44. Barsberg, S.T., Lee, Y.I., and Rasmussen, H.N., *Seed Sci. Res.*, 2018, vol. 28, pp. 41–51.
45. Wang, S., Shen, Q., Su, S., Lin, J., and Song, G., *Trends Chem.*, 2022, vol. 4, pp. 948–961.
46. Song, W., Du, Q., Li, X., Wang, S., and Song, G., *ChemSusChem*., 2022, vol. 15, Article ID: e202200646.
47. Wang, S., Su, S., Xiao, L.P., Wang, B., Sun, R.C., and Song, G., *ACS Sustain, Chem. Eng.*, 2020, vol. 8, pp. 7031–7038.
48. del Río, J.C., Rencoret, J., Gutiérrez, A., Elder, T., Kim, H., and Ralph, J., *ACS Sustain. Chem. Eng.*, 2022, vol. 8, pp. 4997–5012.
49. Li, M., Pu, Y., Meng, X., Chen, F., Dixon, R.A., and Ragauskas, A.J., *Green Chem.*, 2022, vol. 24, pp. 259–270.
50. José, C., Rencoret, J., Gutiérrez, A., Kim, H., and Ralph, J., In: *Advances in Botanical Research*, Sibout, R., Ed., New York, Elsevier, 2022, vol. 104, pp. 1–39.
51. Li, M., Pu, Y., Yoo, C. G., and Ragauskas, A. J., *Green Chem.*, 2016, vol. 18, pp. 1439–1454.
52. Lam, P.Y., Lui, A.C., Wang, L., Liu, H., Umezawa, T., Tobimatsu, Y., and Lo, C., *Front. Plant Sci.*, 2021, vol. 12, Article ID: 733198.
53. Xie, M., Chen, Z., Xia, Y., Lin, M., Li, J., and Lan, W., *Front. Energy Res.*, 2021, vol. 9, Article ID: 756285.
54. del Río, J.C., Rencoret, J., Gutiérrez, A., Lan, W., Kim, H., and Ralph, J., In: *Recent Advances in Polyphenol Research*, Reed, J.D., de Freitas, V.A.P., and Quideau, S., Eds., Hoboken, Wiley Blackwell, 2021, vol. 7, pp. 177–206.
55. Ando, D., Lu, F., Kim, H., Eugene, A., Tobimatsu, Y., and Vanholme, R., *Green Chem.*, 2021, vol. 23, pp. 8995–9013.
56. Rencoret, J., Rosado, M.J., Kim, H., Timokhin, V.I., Gutiérrez, A., and Bausch, F., *Plant Physiol.*, 2022, vol. 188, pp. 208–219.
57. Elder, T., Rencoret, J., Del Río, J.C., Kim, H., and Ralph, J., *Front. Plant Sci.*, 2021, vol. 12, Article ID: 642848. https://doi.org/10.3389/fpls.2021.642848
58. Lawoko, M., Berglund, L., and Johansson, M., *ACS Sustain. Chem. Eng.*, 2021, vol. 9, pp. 5481–5485. https://doi.org/10.1021/acssuschemeng.1c01741
59. Tao, J., Li, S., Ye, F., Zhou, Y., Lei, L., and Zhao, G., *Crit. Rev. Food Sci.*, 2020, vol. 60, pp. 2011–2033. https://doi.org/10.1080/10408398.2019.1625025
60. Lourenço, A. and Pereira, H., In: *Lignin – Trends and Applications*, Poletto, M., Ed., InTechOpen, 2018, pp. 65–98.
61. Yadav, V.K., Gupta, N., Kumar, P., Dashti, M.G., Tirth, V., and Khan, S.H., *Materials*, 2022, vol. 15, Article ID: 953.
62. Balakshin, M., Capanema, E.A., Zhu, X., Sulaeva, I., Potthast, A., Rosenau, T., and Rojas, O.J., *Green Chem.*, 2020, vol. 22, pp. 3985–4001.
63. Su, X., Fu, Y., Shao, Z., Qin, M., Li, X., and Zhang, F., *Ind. Crops Prod.*, 2022, vol. 176, Article ID: 114359.
64. Nadányi, R., Ház, A., Lisý, A., Jablonský, M., Šurina, I., Majová, V., and Baco, A., *Energies*, 2022, vol. 15, Article ID: 6520.
65. Liu, X., Bouxin, F.P., Fan, J., Budarin, V.L., Hu, C., and Clark, J.H., *ChemSusChem*., 2020, vol. 13, pp. 4296–4317.
66. Huang, J., Fu, S., and Gan, L., *Lignin Chemistry and Applications*, Amsterdam, Elsevier, 2019, pp. 25–50.
67. Zhang, H., Zhao, Y., and Zhu, J.K., *Dev. Cell*, 2020, vol. 55, pp. 529–543. https://doi.org/10.1016/j.devcel.2020.10.012
68. Šamec, D., Karalija, E., Šola, I., Vujčić Bok, V., and Salopek-Sondi, B., *Plants*, 2021, vol. 10, Article ID: 118. https://doi.org/10.3390/plants10010118
69. Cesarino, I., *Curr. Opin. Biotechnol.*, 2019, vol. 56, pp. 209–214. https://doi.org/10.1016/j.copbio.2018.12.012
70. Pellegrini, E., Campanella, A., Cotrozzi, L., Tonelli, M., Nali, C., and Lorenzini, G., *Environ. Sci. Pollut. Res.*, 2018, vol. 25, pp. 8148–8160. https://doi.org/10.1007/s11356-017-8818-7
71. Espinosa-Leal, C.A., Mora-Vásquez, S., Puente-Garza, C.A., Alvarez-Sosa, D.S., and García-Lara, S., *Plant Growth Regul.*, 2022, pp. 1–20. https://doi.org/10.1007/s10725-022-00810-3
72. Ninkuu, V., Yan, J., Fu, Z., Yang, T., Ziemah, J., and Ullrich, M.S., *J. Fungi*, 2023, vol. 9, Article ID: 52. https://doi.org/10.3390/jof9010052
73. Chen, K., Guo, Y., Song, M., Liu, L., Xue, H., Dai, H., and Zhang, Z., *Hortic.Res.*, 2020, vol. 7, Article ID: 204. https://doi.org/10.1038/s41438-020-00433-7.
74. Liu, W.G., Hussain, S., Ting, L.I.U., Zou, J.L., Ren, M.L., Tao, Z.H.O.U., *J. Integr. Agric.*, 2019, vol. 18, pp. 43–53. https://doi.org/10.1016/S2095-3119(18)61905-7

75. Lu, Y.C., Lu, Y., and Fan, X., In: *Lignin: Biosynthesis and Transformation for Industrial Applications*, Swati, S. and Kumar, A., Eds., Cham, Springer, 2020, pp. 17–75.
76. Alzagameem, A., Klein, S.E., Bergs, M., Do, X.T., Korte, I., Dohlen, S., Huwe, C., Kreyenschmidt, J., Kamm, B., Larkins, M., and Schulze, M., *Polymers*, 2019, vol. 11, Article ID: 670.
77. Agrawal, R., Kumar, A., Singh, S., and Sharma, K., *J. Polym. Res.*, 2022, vol. 29, Article ID: 222.
78. Espinoza-Acosta, J.L., Torres-Chávez, P.I., Ramírez-Wong, B., López-Saiz, C.M., and Montaño-Leyva, B., *BioResources*, 2016, vol. 11, pp. 5452–5481.
79. *Lignin and Lignans: Advances in Chemistry*, Heitner, C., Dimmel, D., and Schmidt, J., Eds., Boca Raton, CRC, 2016.
80. *Polyphenols: Mechanisms of Action in Human Health and Disease*, Watson, R.R., Preedy, V.R., and Zibadi, S., Eds., 2nd ed., New York, Academic, vol. 1, 2018.
81. Vinardell, M.P and Mitjans, M., *Int. J. Mol. Sci.*, 2017, vol. 18, pp. 1–15. https://doi.org/10.3390/ijms18061219
82. Singh, S.K., *Int. J. Biol. Macromol.*, 2019, vol. 132, pp. 265–277. https://doi.org/10.1016/j.ijbiomac.2019.03.182
83. Sato, S., Mukai, Y., Yamate, J., Norikura, T., Morinaga, Y., Mikame, K., Funaoka, M., and Fujita, S., *Free Radical Res.*, 2009, vol. 43, pp. 1205–1213. https://doi.org/10.3109/10715760903247264
84. Saluja, B., Thakkar, J.N., Li, H., Desai, U.R., and Sakagami, M., *Pulm. Pharmacol. Ther.*, 2013, vol. 26, pp. 296–3043. https://doi.org/10.1016/j.pupt.2012.12.009
85. Mehta, A.Y., Mohammed, B.M., Martin, E.J., Brophy, D.F., Gailani, D., and Desai, U.R., *J. Thromb. Haemost.*, 2016, vol. 14, pp. 828–838.
86. Revajová, V., Levkut, M., Levkutová, M., Bořutová, R., Grešaková, Ľ., Košiková, B., and Leng, Ľ., *Acta Vet. Hung.*, 2013, vol. 61, pp. 354–365.
87. Heiss, C., Spyridopoulos, I., and Haendeler, J., *Exp. Gerontol.*, 2018, vol. 109, pp. 108–118. https://doi.org/10.1016/j.exger.2017.06.015
88. Borisenkov, M.F., Karmanov, A.P., and Kocheva, L.S., *Usp. Gerontol.*, 2005, no. 17, pp. 34–41.
89. Hudlikar, R.R., Venkadakrishnan, V.B., Kumar, R., Thorat, R.A., Kannan, S., Ingle, A.D., Desai, S., Maru, G.B., and Mahimkar, M.B., *Mol. Carcinog.*, 2017, vol. 56, pp. 625–640. https://doi.org/10.1002/ mc.22521
90. Gonzalez de Mejia, E. and Loarca-Pina, G., *Food Res. Int.*, 2018, vol. 105, pp. 159–168. https://doi.org/10.1016/j.foodres.2017.11.004
91. Liu, D., He, B., Lin, L., Malhotra, A., and Yuan, N., *Drug Chem. Toxicol.*, 2019, vol. 42, pp. 328–334. https://doi.org/10.1080/01480545.2018.1523921
92. Figueiredo, P., Lintinen, K., Kiriazis, A., Hynninen, V., Liu, Z., and Bauleth-Ramos, T., *Biomaterials*, 2017, vol. 121, pp. 97–108. https://doi.org/10.1016/j.biomaterials.2016.12.034
93. Panchenko, A.V., Fedoros, E.I., Pigarev, S.E., Maydin, M.A., Gubareva, E.A., Kireeva, G.S., Tyndyk, M.L., Kuznetsova, A.I., Nekhaeva, T.L., Danilova, A.B., Baldueva, I.A., and Anisimov, V.N., *Integr. Cancer Ther.*, 2019, vol. 18, pp. 1–13. https://doi.org/10.1177/1534735419833778
94. Saratale, R.G., Saratale, G.D., Ghodake, G., Cho, S.-K., Kadam, A., Kumar, G., Jeon, S.-H., Pant, D., Bhatnagar, A., and Shin, H.S., *Int. J. Biol. Macromol.*, 2019, vol. 128, pp. 391–400. https://doi.org/10.1016/j.ijbiomac.2019.01.120
95. Schneider, W.D.H., Dillon, A.J.P., and Camassola, M., *Biotechnol. Adv.*, 2021, vol. 47, Article ID: 107685. https://doi.org/10.1016/j.biotechadv.2020.107685
96. Verdini, F., Calcio Gaudino, E., Canova, E., Tabasso, S., Jafari Behbahani, P., and Cravotto, G., *Molecules*, 2022, vol. 27, Article ID: 3598. https://doi.org/10.3390/molecules27113598
97. Ali, D.A. and Mehanna, M.M., *Int. J. Biol. Macromol.*, 2022, vol. 221, pp. 934–953. https://doi.org/10.1016/j.ijbiomac.2022.09.007
98. Liu, R., Dai, L., Xu, C., Wang, K., Si, C., and Zheng, C., *ChemSusChem*., 2020, vol. 13, pp. 4266–4283. https://doi.org/10.1002/cssc.202000783
99. Sipponen, M.H., Lange, H., Crestini, C., Henn, A., and Österberg, M., *ChemSusChem*., 2019, vol. 12, pp. 2039–2054. https://doi.org/10.1002/cssc.201900480
100. Figueiredo, P., Lintinen, K., Hirvonen, J.T., Kostiainen, M.A., and Santos, H.A., *Prog. Mater. Sci.*, 2018, vol. 93, pp. 233–269.
101. Xu, J., Xu, J.J., Lin, Q., Jiang, L., Zhang, D., and Li, Z., *ACS Appl. Bio Mater.*, 2020, vol. 4, pp. 3–13. https://doi.org/10.1021/acsabm.0c00858
102. Roopan, S.M., *Int. J. Biol. Macromol.*, 2017, vol. 103, pp. 508–514. https://doi.org/10.1016/j.ijbiomac.2017.05.10

103. Li, S.-X., Li, M.-F., Bing, J., Wu, X.-F., Peng, F., and Ma, M.-G., *Int. J. Biol. Macromol.*, 2019, vol. 132, pp. 836–843. https://doi.org/10.1016/j.ijbiomac.2019.03.17
104. Sunthornvarabhas, J., Rungthaworn, P., Sukatta, U., Juntratip, N., and Sriroth, K., *Sugar Tech.*, 2020, vol. 22, pp. 697–705. https://doi.org/10.1007/s12355-019-00778-x
105. Lobo, F.C.M., Franco, A.R., Fernandes, E.M., and Reis, R.L., *Molecules*, 2021, vol. 26, Article ID: 1749.
106. Gujjala, L.S., Kim, J., and Won, W., *J. Cleaner Prod.*, 2022, vol. 363, Article ID: 132585. https://doi.org/10.1016/j.jclepro.2022.132585
107. Chen, M., Li, Y., Liu, H., Zhang, D., Shi, Q.-S., Zhong, X.-Q., Guo, Y., and Xie, X.-B., *Mater. Today Bio*., 2022, vol. 18, Article ID: 100520. https://doi.org/10.1016/j.mtbio.2022.100520
108. Moreira, W.M., Moreira, P.V.V., Dos Santos, D.F., Gimenes, M.L., and Vieira, M.G.A., *Environ. Sci. Pollut. Res.*, 2023, vol. 30, pp. 19564–19591. https://doi.org/10.1007/s11356-023-25150-1
109. Raghuraman, M., Verma, P., Kunwar, A., Phadnis, P.P., Jain, V.K., and Priyadarsini, I., *Metallomics*, 2017, vol. 9, pp. 715–725. https://doi.org/10.1039/c7mt00034k
110. Srisapoome, P., Hamano, K., Tsutsui, I., and Iiyama, K., *Fish Shellfish Immunol.*, 2018, vol. 72, pp. 494–501. https://doi.org/10.1016/j.fsi.2017.11.037
111. Gordts, S.C., Férir, G., D'huys, T., Petrova, M.I., Lebeer, S., Snoeck, R., Andrei, G., and Schols, D., *PLoS One*, 2015, vol. 10, Article ID: e0131219. https://doi.org/10.1371/journal.pone.0131219
112. Lee, J.B., Yamagishi, C., Hayashi, K., and Hayashi, T., *Biosci., Biotechnol. Biochem.*, 2011, vol. 75, pp. 459–465. https://doi.org/10.1271/bbb.100645
113. Yang, W., Fortunati, E., Dominici, F., Giovanale, G., Mazzaglia, A., Balestra, G.M., Kenny, J.M., and Puglia, D., *Eur. Polym. J.*, 2016, vol. 79, pp. 1–12. https://doi.org/10.1016/j.eurpolymj.2016.04.003
114. Cazacu, G., Capraru, M., and Popa, V., *Advances in Natural Polymers*, Berlin: Springer, 2013, vol. 18, pp. 255–312.
115. Yang, W., Fortunati, E., Dominici, F., Giovanale, G., Mazzaglia, A., Balestra, G.M., Kenny, J.M., and Puglia, D., *Int. J. Biol. Macromol.*, 2016, vol. 89, pp. 360–368. https://doi.org/10.1016/j.ijbiomac.2016.04.068
116. Andrianova, E.N., *Ptitsevodstvo*, 2017, no. 6, pp. 13–16.
117. Klapiszewski, L. and Jesionowski, T., Thakur, V.K., Thakur, M.K., and Kessler, M.R., Eds., *Handbook of Composites from Renewable Materials*, Hoboken: Wiley, 2017, pp. 519–554. https://doi.org/10.1002/9781119441632.ch122
118. Golovnya, E.Ya., *Zhivotnov. Rossii*, 2021, pp. 38–39.
119. Borisenkov, M.F., Karmanov, A.P., Kocheva, L.S., Markov, P.A., Istomina, E.I., Bakutova, L.A., Litvinets, S.G., Martinson, E.A., Durnev, E.A., Vityazev, F.A., and Popov, S.V., *Int. J. Polymer. Mater. Polymer. Biomater.*, 2016, vol. 65, pp. 433–441. https://doi.org/10.1080/00914037.2015.1129955
120. Solihat, N., Sari, F., Falah, F., Ismayati, M., Lubis, M.A.R., Fatriasari, W., Santoso, E., and Syafii, W., *J. Sylva Lest.*, 2021, vol. 9, pp. 1–22. https://doi.org/10.23960/jsl191-22
121. Xie, D., Gan, T., Su, C., Han, Y., Liu, Z., and Cao, Y., *Int. J. Biol. Macromol*., 2020, vol. 161, pp. 315–324. https://doi.org/10.1016/j.ijbiomac.2020.06.049
122. Piccinino, D., Capecchi, E., Tomaino, E., Gabellone, S., Gigli, V., Avitabile, D., and Saladino, R., *Antioxidants*, 2021, vol. 10, Article ID: 274. https://doi.org/10.3390/antiox10020274
123. Su, C., Gan, T., Liu, Z., Chen, Y., Zhou, Q., Xia, J., and Cao, Y., *Int. J. Biol. Macromol.*, 2021, vol. 184, pp. 369–379. https://doi.org/10.1016/j.ijbiomac.2021.06.063
124. Hsu, B., Coupar, I.M., and Ng, K., *Food Chem.*, 2006, vol. 98, pp. 317–328. https://doi.org/10.1016/j.foodchem.2005.05.077
125. Dong, X., Dong, M., Lu, Y., Turley, A., Jin, T., and Wu, C., *Ind. Crops Prod.*, 2011, vol. 34, pp. 1629–1634. https://doi.org/10.1016/j.indcrop.2011.06.002
126. Lapin, A.A., Borisenkov, M.F., Karmanov, A.P., Berdnik, I.V., Kocheva, L.S., Musin, R.Z., and Magdeev, I.M., *Khim. Rast. Syr'ya,* 2007, pp. 79–83.
127. Kocheva, L.S., Borisenkov, M.F., Karmanov, A.P., and Zagirova, S.V., RF Patent 2292896, 2007.
128. Kocheva, L.S., Karmanov, A.P., and Borisenkov, M.F., *Usp. Gerontol.*, 2008, vol. 21, pp. 494–495.
129. Kumar, R., Butreddy, A., Kommineni, N., Reddy, P.G., Bunekar, N., Sarkar, C., Dutt, S., Mishra, V.K., Aadil, K.R., Mishra, Y.K., Oupicky, D., and Kaushik, A., *Int. J. Nanomed.*, 2021, vol. 16, pp. 2419–2441. https://doi.org/10.2147/IJN.S303462
130. Wardman, P., *Brit. J. Radiol.*, 2009, vol. 82, pp. 89–104. https://doi.org/10.1259/bjr/60186130

131. Gudkov, S.V., Popova, N.R., and Bruskov, V.I., *Biophysics*, 2015, vol. 60, pp. 659–667. https://doi.org/10.1134/S0006350915040120
132. Hosseinimehr, S.J., *Drug Discov. Today*, 2007, vol. 12, pp. 794–805. https://doi.org/10.1016/j.drudis.2007.07.017
133. Panchenko, A.V., Fedoros, E.I., Pigarev, S.E., Bykov, V.N., Drachev, I.S., and Kraev, S.Yu., *Radiats. Biol. Radioekol.*, 2017, vol. 57, pp. 1–7. https://doi.org/10.7868/S086980311705006X
134. Bykov, V.N., Drachev, I.S., Kraev, S.Y., Maydin, M.A., Gubareva, E.A., Pigarev, S.E., Anisimov, V.N., Baldueva, I.A., Fedoros, E.I., and Panchenko, A.V., *Int. J. Radiat. Biol.*, 2018, vol. 94, pp. 114–123. https://doi.org/10.1080/09553002.2018.1416204
135. Bykov, V.N., Drachev, I.S., Panchenko, A.V., Fedoros, E.I., and Pigarev, S.E., *Radiats. Biol. Radioekol.*, 2020, vol. 60, pp. 404–410. https://doi.org/10.31857/S0869803120040050
136. Kuntic, V.S., Stankovic, M.B., Vujic, Z.B., Brboric, J.S., and Uskokovic-Markovic, S.M., *Chem. Biodivers.*, 2013, vol. 10, pp. 1791–1803. https://doi.org/10.1002/cbdv.201300054
137. Qazi, S.S., Li, D., Briens, C., Berruti, F., and Abou-Zaid, M., *Molecules*, 2017, vol. 22, Article ID: 372. https://doi.org/10.3390/molecules22030372
138. Pigarev, S.E., Trashkov, A.P., Panchenko, A.V., Yurova, M.N., Bykov, V.N., Fedoros, E.I., and Anisimov, V.N., *Environ. Res.*, 2021, vol. 192, Article ID: 110321. https://doi.org/10.1016/j.envres.2020.110321
139. Azqueta, A. and Collins, A., *Nutrients*, 2016, vol. 8, Article ID: 785. https://doi.org/10.3390/nu8120785
140. Labaj, J., Slamenova, D., and Kosikova, B., *Nutr. Cancer*, 2003, vol. 47, pp. 95–103. https://doi.org/10.1207/s15327914nc4701_12
141. Wang, Y., Duan, H., Dai, Y., Bin, P., Cheng, J., Pan, Z., Huang, C., Leng, S., and Zheng, Y., *Sci. Total Environ.*, 2009, vol. 407, pp. 2615–2620. https://doi.org/10.1016/j.scitotenv.2009.01.002
142. Assimopoulou, A.N., Sinakos, Z., and Papageorgiou, V.P., *Phytother. Res.*, 2005, vol. 19, pp. 997–1000. https://doi.org/10.1002/ptr.1749
143. Papież, M.A., *Drug Chem. Toxicol.*, 2013, vol. 36, pp. 93–101. https://doi.org/10.3109/01480545.2012.726626
144. Kuhlmann, M.K., Horsch, E., Burkhardt, G., Wagner, M., and Köhler, H., *Arch. Toxicol.*, 1998, vol. 72, pp. 536–540. https://doi.org/10.1007/s002040050539
145. Barsberg, S., Elder, T., and Felby, C., *Chem. Mater.*, 2003, vol. 15, pp. 649–655. https://doi.org/10.1021/cm021162s
146. Bykov, V.N., *Usp. Gerontol.*, 2020, vol. 33, pp. 646–656. https://doi.org/10.34922/AE.2020.33.4.003
147. Belyi, V.A., Pechnikova, A.A., Kocheva, L.S., Moskalev, A.A., and Karmanov, A.P., *Usp. Gerontol.*, 2010, vol. 23, pp. 221–227.
148. Moskalev, A., Chernyagina, E., de Magalhaes, J.P., Barardo, D., Thoppil, H., Shaposhnikov, M., Budovsky, A., Fraifeld, V.E., Garazha, A., Tsvetkov, V., Bronovitsky, E., Bogomolov, V., Scerbacov, A., Kuryan, O., Gurinovich, R., Jellen, L.C., Kennedy, B., Mamoshina, P., Dobrovolskaya, E., Aliper, A., Kaminsky, D., and Zhavoronkov, A., *Aging*, 2015, vol. 7, pp. 616–628. https://doi.org/10.18632/aging.100799
149. Fedoros, E.I., Badun, G.A., Grozdova, I.D., Melik-Nubarov, N.S., Perminova, I.V., Baldueva, I.A., Danilova, A.B., Nehaeva, T.L., Ryakhovskiy, A.A., Pigarev, S.E., Panchenko, A.V., and Anisimov, V.N., In: Book of Abstracts: *Fifth International Conference of CIS IHSS on Humic Innovative Technologies "Humic Substances and Living Systems" (HIT-2019)*, October 19–23, 2019, Moscow, 2019, p. 101. https://doi.org/10.36291/HIT.2019.fedoros.089

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